

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 2

6031/2
1 hour 30 minutes

NOVEMBER 2024 SESSION

Candidates answer on the question paper.
Additional materials:
Data Booklet
Electronic Calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
6	
TOTAL	

This question paper consists of 10 printed pages.

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Answer all questions

Write your answers in the spaces provided on the question paper.

- 1 (a) Potassium chlorate (I) decomposed slowly at a temperature of 75°C to form two chlorine containing compounds.

(i) Write a chemical equation for the decomposition of potassium chlorate (I).

(ii) Name the type of reaction in (i).

[2]

- (b) Fig.1.1 shows how the concentration of potassium chlorate (I) changed with time during the decomposition.

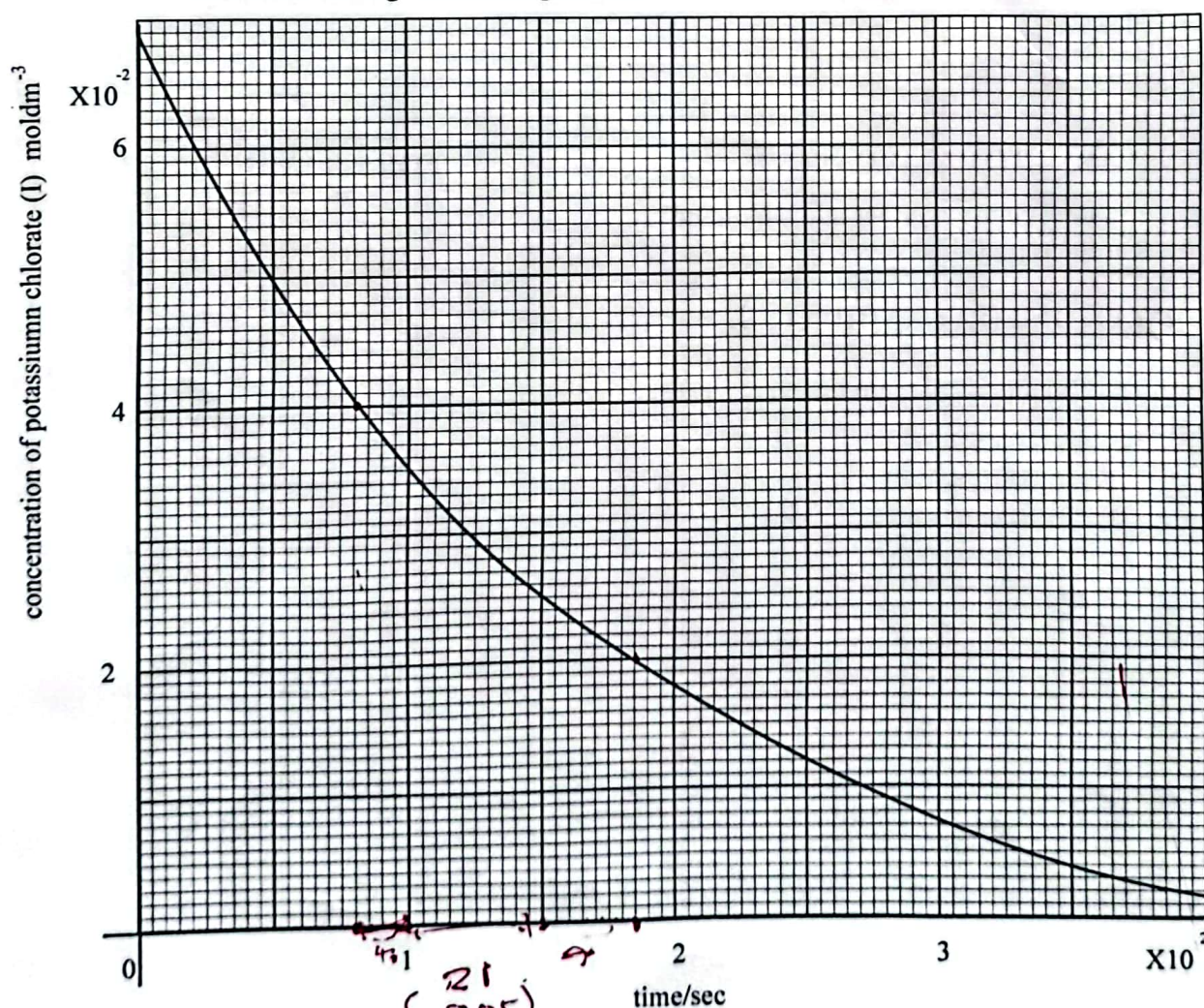


Fig. 1.1

- (i) Deduce, with a reason, the order of reaction with respect to potassium chlorate (I).

- (ii) Write the rate equation for the reaction.

- (iii) Calculate the value for the rate constant, K .

- (iv) Determine the rate of reaction when the concentration of potassium chlorate (I) is 0.02 mol dm^{-3} .

[5]

- (c) (i) Oxygen exists in two forms, dioxygen, O_2 , and trioxygen, O_3 .

Give the term used to describe the two forms of oxygen element.

- (ii) Describe how chlorofluorocarbons, CFCs, lead to damage of the ozone layer.

[3]

[Total:10]

- 2 (a) Fe^{3+} ions react with I^- ions according to the equation.



- (i) State, with a reason, the type of reaction undergone by Fe^{3+} ions.

- (ii) Calculate the cell voltage for the reaction.

[3]

- (b) Fig. 2.1 shows an electrochemical cell, set up by a student to determine the standard potential for $\text{Cu}^{2+}(\text{aq})/\text{Cu}(\text{s})$.

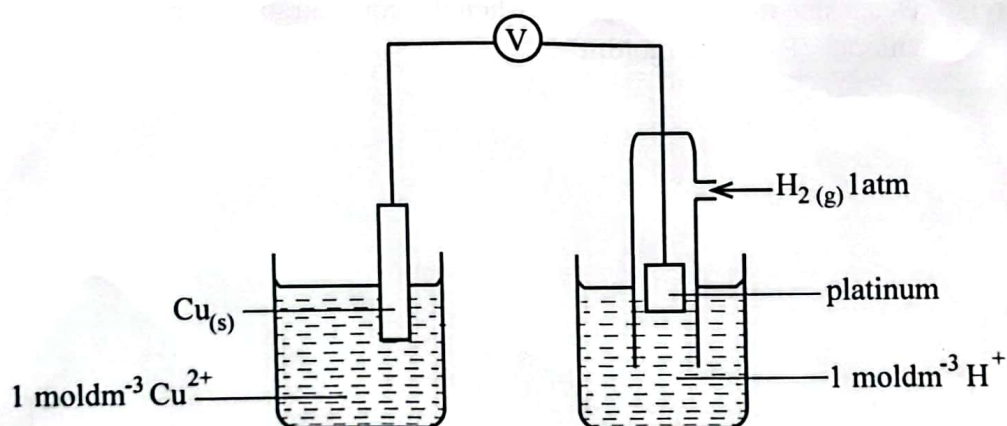


Fig. 2.1

- (i) Explain why the cell will **not** work.

- (ii) Identify **one** other condition that should be given for the reading on the voltmeter to be considered as the accurate E^θ for $\text{Cu}^{2+}(\text{aq})/\text{Cu}(\text{s})$ cell.

[2]

(c) Chlorine is produced by the electrolysis of brine using the diaphragm cell.

(i) Give the equation for the reaction at the cathode.

(ii) Explain why titanium is used as the anode.

[2]

(d) (i) Give **one** application of DNA fingerprinting.

(ii) Explain why all DNA fragments move towards the positive electrode in electrophoresis.

[3]

[Total:10]

- 3 (a) Define the term *periodicity*.

[1]

- (b) Complete Table 3.1 by writing the formula of each chloride and the oxidation state of the elements shown.

Table 3.1

	phosphorus	sulphur
formula of chloride		
oxidation state		

[2]

- (c) State the acid-base nature of the solutions formed when sodium chloride and silicon tetrachloride are separately dissolved in water.

sodium chloride _____

silicon tetrachloride _____

[2]

- (d) (i) Write a balanced chemical equation for the reaction between concentrated sulphuric acid and sodium chloride.

[2]

- (ii) State, with a reason, **one** precaution to be considered when carrying out the experiment in (d) (i).

[3]

[Total:10]

- 4 (a) (i) State the observable changes that occur when ethanal reacts with alkaline copper(II) sulphate solution.
- _____
- (i) Construct balanced half equations for the reaction between ethanal and alkaline copper (II) sulphate.
- _____
- _____ [3]
- (b) Outline any **three** differences between addition and condensation polymerisation.
- 1 _____
- 2 _____
- 3 _____ [3]
- (c) (i) Give the structure of the shortest secondary alcohol that exhibits optical isomerism.
- (ii) Draw the displayed formula of the optical isomers.

- (ii) Potassium hydroxide reacts with a haloalkane to produce an alkene.

Explain the use of ethanol in the reaction.

_____ [4]

[Total:10]

- 5 (a) (i) Define the term *nucleophile*.

_____ [1]

- (ii) A ketone of molecular formula C_4H_8O , reacts with hydrogen cyanide in the presence of NaOH to form an organic compound T.

Describe the mechanism for the reaction.

[4]

- (iii) Explain the importance of this type of reaction in organic synthesis.

_____ [2]

- (b) Table 5.1 shows pK_a values of three organic acids.

Table 5.1

acid	pK_a value
ethanoic acid	4.77
bromoethanoic acid	2.90
chloroethanoic acid	2.89

Explain the trend in the pK_a values of the three acids.

_____ [3]
[Total: 10]

- 6 (a) State one industrial method of controlling sulphur dioxide emission.

_____ [1]

- (b) A solution of acidified copper (I) chloride, $\text{CuCl}_{(\text{aq})}$, disproportionates when diluted with water to form a reddish-orange solid and a light blue solution.

- (i) State the colour of $\text{CuCl}_{(\text{aq})}$.

- (ii) Identify the

1 reddish orange solid, _____

2 light blue solution. _____

- (iii) Write an equation for the disproportionation of $\text{CuCl}_{(\text{aq})}$.

[4]

- (c) Fig. 6.1 shows the chromatogram obtained from a two-way paper chromatography of a mixture of amino acids A, B, C and D.

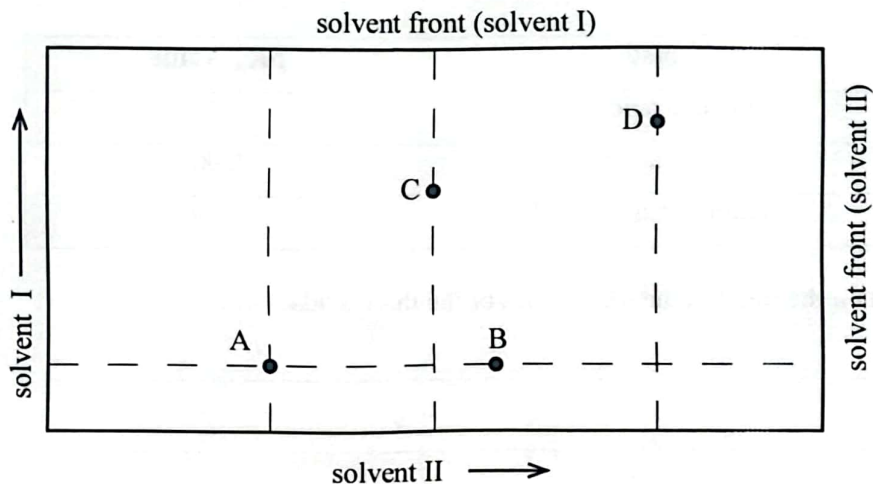


Fig. 6.1

- (i) Name a reagent that can be used to locate the amino acids.
- _____
- (ii) Identify the **two** amino acids which were not separated by use of solvent I.
- _____
- (iii) Explain the relative positions of the amino acids on the chromatogram.

[5]
[Total:10]